1.5.5 ▶ ▶ ▶ Shell Theorem

Newton's law of universal gravitation provides an easy way to calculate the gravitational force between two point-like particles, but not for continuous bodies. Newton's Shell Theorem provides a way to calculate the gravitational field emanating from a spherical shell or solid sphere by integrating over small point-like masses, solving the issue of not knowing the gravitational field instated by continuous bodies.

To derive the Shell Theorem, consider a point at a distance D from a spherical shell of mass M and radius R. Furthermore, consider a slice of the shell as described by the following diagram

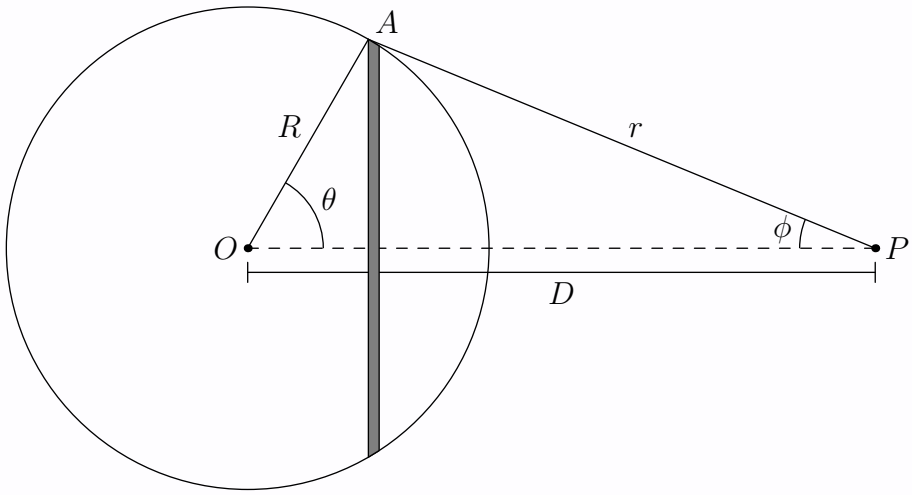

Fig. 1.14

At point P, the gravitational field due to the slice dg is given by

$$dg = G\frac{dm}{r^2}\cos\phi \tag{1.80}$$

where dm is the mass of the slice and only the horizontal component is taken since the vertical components from each point on the slice cancel out. The mass of the slice is given by

$$dm = \sigma 2\pi R \sin\theta R d\theta$$

where $\sigma = \frac{M}{4\pi R^2}$ is the surface mass density of the shell. Plugging values into eq. 1.76 and rearranging yields

$$dg = \frac{GM}{2}\cdot\frac{\sin\theta d\theta}{r^2}\cos\phi. \tag{1.81}$$

To simplify eq. 1.80, we can write two laws of cosines:

$$r^2 = R^2 + D^2 - 2RD\cos\theta$$

$$R^2 = r^2 + D^2 - 2rD\cos\phi$$

Differentiating the first one yields the relation

$$rdr = RD\sin\theta d\theta$$

Substituting all values into eq. 1.80 to write things in terms of r results in the expression

$$\begin{aligned} dg &= \frac{GM}{4RD^2} \cdot \frac{r^2 + D^2 - R^2}{r^2} dr \\ &= \frac{GM}{4RD^2} \left(1 + \frac{D^2 - R^2}{r^2}\right) dr \end{aligned} \tag{1.82}$$

Since r changes from $D - R$ to $D + R$ as it moves across the sphere, we integrate over those bounds

$$\begin{aligned} g &= \frac{GM}{4RD^2} \int_{D-R}^{D+R} 1 + \frac{D^2 - R^2}{r^2} dr \\ &= \frac{GM}{4RD^2} \left[r - \frac{D^2 - R^2}{r} \right]_{D-R}^{D+R} \\ &= \frac{GM}{4RD^2} \left[D + R - (D - R) - (D - R - (D + R)) \right] \\ &= \frac{GM}{4RD^2} [4R] = \boxed{\frac{GM}{4D^2}}. \end{aligned} \tag{1.83}$$

This result states that we can simply treat the spherical shell as a point mass located at the center of the sphere for points outside of the sphere. What if we were inside the sphere, though? Consider the following diagram, which is identical to Fig. 1.13 except point P is inside the shell.

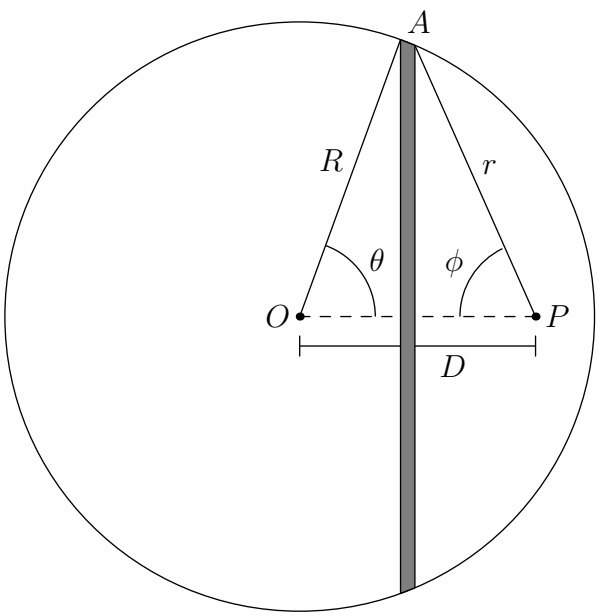

Fig. 1.15

In this case, the expression for the gravitational field at P is the same as the previous case, except the bounds of integration now go from $R - D$ to $R + D$. Calculating the

gravitational field at P using these new bounds gives us

$$\begin{aligned} g &= \frac{GM}{4RD^2}\left[r + \frac{R^2 - D^2}{r}\right]_{R-D}^{R+D} \\ &= \frac{GM}{4RD^2}\left[R - D + R + D - (R + D + R - D)\right] \\ &= \frac{GM}{4RD^2}\left[0\right] = \boxed{0}. \end{aligned} \tag{1.84}$$

This result that there is no gravitational field inside of a spherical shell can be surprising at first, but one can intuitively make sense of it by drawing a vertical line through P, and considering that although more mass lies to the left side of the line, it is on average farther away from P and vice versa[11].

Similar to the technique used in 1.4.3, we can now extend the result from eq. 1.83 to a solid sphere. For a point at a distance D from the center of a uniform solid sphere with radius R and mass M, the gravitational field due to a thin shell of the sphere is

$$dg = \frac{G\rho}{D^2}4\pi r^2 dr = \frac{3GM}{D^2R^3}r^2 dr.$$

Where $\rho = \frac{3M}{4\pi R^3}$ is the volume mass density of the sphere. Integrating over the entire sphere yields

$$g = \boxed{\frac{GM}{D^2}}, \tag{1.85}$$

which makes sense, since each shell of the sphere can be treated as a mass located at the center of the sphere.

To summarize, the gravitational field inside a spherical shell is zero, the gravitational field outside of solid spheres and spherical shells can be calculated by treating the sphere as a mass located at the center of the sphere ★

[11] Don't take this explanation too literally, it is just a way to make sense of the result!